

CSE 3550/5000: Blockchain Technology

Lecture 10 **Wallets and Key Management**

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Outline

- Key management in cryptocurrencies.
 - Hot and cold storage.
 - Hierarchical wallets.
 - Cold info storage.
 - Splitting and sharing keys.
 - Online wallets.

Spending Coins

- Recall that coins in any cryptocurrency are virtual.
 - Strings of bits.
- Spending any amount of coins requires:
 - Some public information from the blockchain.
 - The secret key associated with the address (or public key) that owns the coins.
 - To produce a digital signature proving ownership of coins.
- Losing the secret keys means losing these coins; no one will be able to spend them.

Key Management

- Storage and retrieval of secret keys.
 - Involves also when and how to generate new keys for future transactions.
- Goals:
 - Security; only the legitimate owner gets to spend a given coin.
 - Availability; owners can spend their coins whenever they wish.
 - Usability; it is relatively easy for the average user to store/retrieve/use secret keys.
- We will focus on Bitcoin in the slides.
 - The concepts we will study can be applied to any other cryptocurrency.

Wallet Software

- User (or client) software that keeps track of coins that a client owns.
- Provides a convenient user interface to simplify operations.
 - Issuing transactions.
 - Tracking total balance.
 - Generate keys when needed.
 - Bookkeeping of these keys.
- Encode addresses as text strings (Base58) or in the form of QR codes.
 - Simplifies sharing addresses with others.
- A large number of wallets is available.
 - Different flavors; desktop, mobile, web applications, etc.
 - Different vendors; metamask, coinbase, etc.
 - Security is a driving factor for which one to choose.

Naive Solution

- Store the keys in a file on your laptop or smartphone.
 - Security is tied to your device; breaking into the device allows an attacker to steal your keys (and so your coins).
- This is called hot storage.
 - Easy to use but risky.

Hot vs. Cold Storage I

- Hot storage.
 - Storing secret keys on a device that is connected to the Internet and used frequently for variety of applications.
 - Usable/covenient.
- Cold storage.
 - Offline storage, like on a machine or memory device that is stored in some safe location.
 - Less convenient.
- Good practice: The majority of the coins are in the cold storage with a few in the hot storage.



Hot vs. Cold Storage II

- Separate keys are needed for each.
 - Otherwise, compromising hot storage will compromise cold storage.
- But both should be aware of their addresses to allow transferring currency.
 - Cold storage stores its cold secret keys, cold addresses, and hot public addresses.
 - Hot storage stores hot secret keys and public addresses, as well as cold public addresses.

New Addresses for Cold Storage?

- To break transaction linkability, a good practice in Bitcoin is to create a new address for each new transactions.
- How can a hot wallet learn the addresses of a cold wallet?
 - Remember cold wallet is offline, no internet connectivity.
- Even generating a large chunk of addresses at the beginning will not work.
 - Cold storage needs to connect whenever a new batch of addresses is generated.

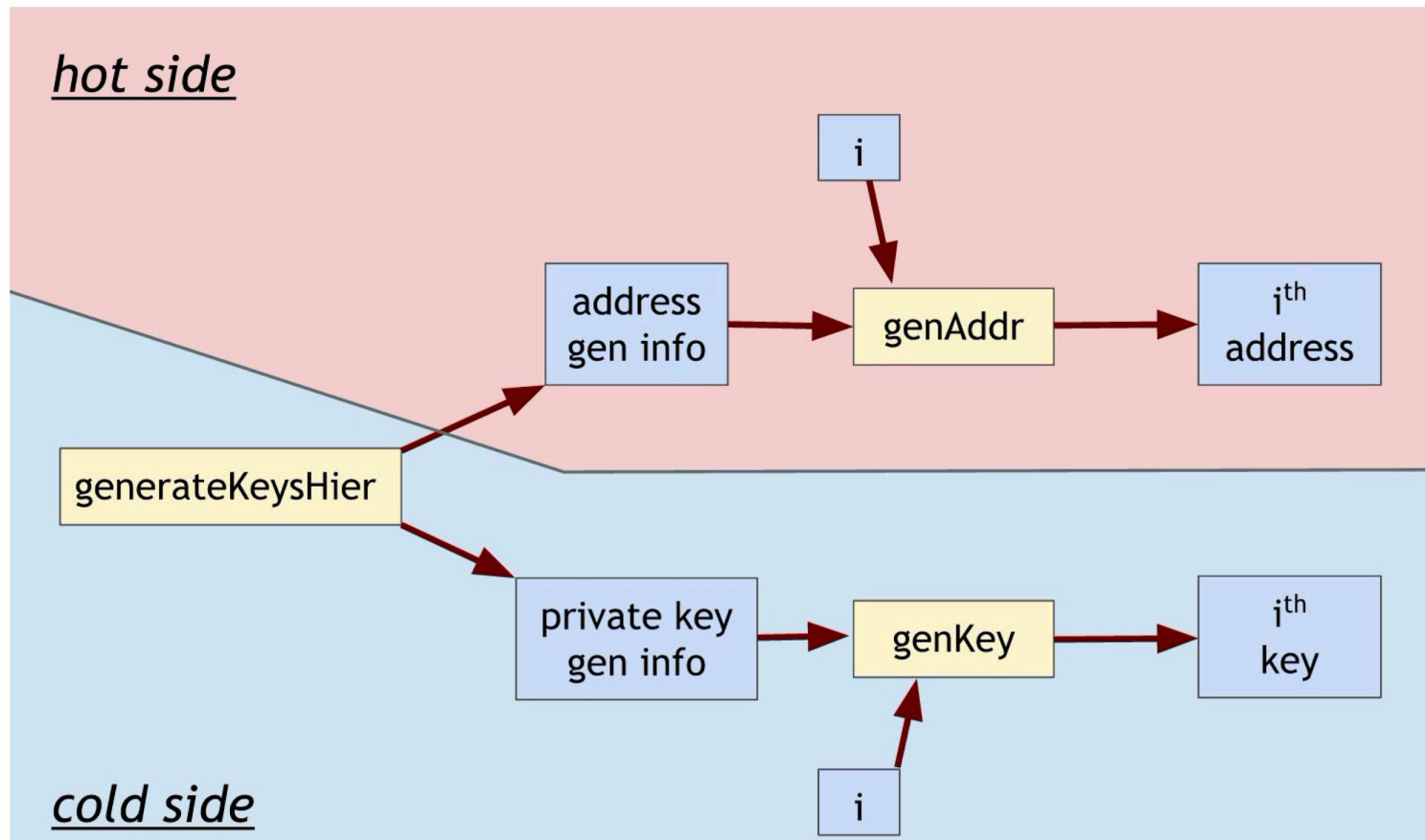
Hierarchical Key Generation I

- Allows cold storage to generate an infinite number of key pairs (or addresses).
- Usual keyGen algorithms generate a single key pair; public and private.
- Hierarchical keyGen instead generates some info that allows generating all future keys in a deterministic way.
 - Usually called master public key and master secret key.
- The master public key does not reveal any info about the master secret key or all future generated secret keys.

Hierarchical Key Generation II

- By having the operation indexed with some integer i , the master key allows generating the i^{th} future key.
 - Works for both public and secret keys.
- Hot storage has a copy of the master public key, while the master secret key is known only to the cold storage.
- Not all digital signature schemes support this hierarchical approach.
 - ECDSA supports that, details come later.

Hierarchical KeyGen - Pictorially



*From Chapter 5, Bitcoin and Cryptocurrency Technologies book.

Hierarchical KeyGen in ECDSA (BIP32)

- A group G , in which DDH is believed to be hard, of order p (where p is some large prime number) and a group generator g .
- KeyGen is extended into 3 algorithms (H below is a hash function modeled as a random oracle):
 - $\text{keyGenHer}(1^n)$: $\text{msk} = x$, $\text{mpk} = g^x$, Hash function H .
 - n is the security parameter.
 - x is some integer selected at random from $\mathbb{Z}_p$.
 - $\text{addrGen}(\text{mpk}, i)$: $r = H(i \parallel \text{mpk})$, $\text{pk}_i = \text{mpk} \cdot g^r = g^{x+r}$, $\text{addr}_i = H(\text{pk}_i)$
 - $\text{keyGen}(\text{msk}, i)$: $r = H(i \parallel \text{mpk})$, $\text{sk}_i = \text{msk} + r = x+r$
- A hot wallet will be able to generate the i th address (or public key) and a cold storage will be able to generate the i th private key without any interaction.

Security Issue in the Previous Scheme

- No forward secrecy.
 - Given i , ski , mpk , it is easy to determine msk .
 - Assume an attacker compromised the hot wallet (got hold of mpk), and one secret key of the cold storage has been leaked.
 - This allows the attacker to compute msk , and hence, all previous as well as future secret keys.
 - ***Do you see why?***
- Source of vulnerability; the way the randomness r is computed.
- Many modifications were proposed to fix this problem.
 - Out of scope.
- This shows how proposing security countermeasure is not an easy task!
 - Put differently, restricting the attacker capability is not the way to go.

Cold Info Storage

- On a device, and lock that device in a safe.
- Brain wallets:
 - Encrypt under a password that the user memorizes.
 - Security issue: offline password cracking.
- Paper wallet:
 - Print the key material and store the paper in a safe.
- Tamper resistant hardware device.
 - Either the device generates the secret key and stores it, or just use it to store the private key.
 - The device signs the transactions; it does not allow retrieving the secret key at all.

Splitting and Sharing Keys

- Distribute the trust of storing a key among multiple devices/locations instead of one.
- Works by creating shares of the key and store each share at a different place (or with a different party).
 - Usually a threshold is used, meaning that having t out of n shares allows constructing the secret key.
- Good for both availability and security.
 - Any t shares can be used.
 - As long as less than t shares are revealed, no information will be leaked about the secret key.
 - Much better than having a single key stored at a single location.

Additive Secret Sharing

- It is a full threshold scheme (meaning all shares are needed to reconstruct the secret).
- Example 2 out of 2 or (2,2)-AddSS (all operations are integers within some finite group):
 - **Share:** pick a random value as the first share, so $s_1 = r$, then set the second share to be $s_2 = s - r$.
 - **Reconstruct:** given s_1 and s_2 , compute $s = s_1 + s_2$
- This can be extended to any full threshold (3,3), (4,4), ...
- Full threshold secret sharing distribute trust but lacks robustness.
 - If a single share is lost, the whole secret is lost.
- Threshold secret sharing (t-out-of-n secret sharing) using, e.g., Shamir secret sharing (see next) supports both trust distributions and robustness.

Shamir Secret Sharing I

- Supports any threshold t out of n .
- The basic idea is:
 - Generate a random polynomial of degree $(t-1)$ with the free coefficient set as the secret key.
 - Each share is simply the evaluation of the polynomial at the share index.
 - Reconstruct is computing the free coefficient by applying lagrange interpolation over t shares.
- Example: $(3,5)$ -SSS over Z_{11}
 - $f(x) = S + a_1x + a_2x^2 \pmod{11}$, shares are $(i, f(i))$, and the secret is S
 - Say $S = 4$, choose $f(x) = 4 + 2x + 3x^2 \pmod{11}$
 - Shares are $(1, 9), (2, 9), (3, 4), (4, 5), (5, 1)$
 - Any 3 shares uniquely define the original polynomial, and so the secret which is $f(0)$.

Shamir Secret Sharing II

- So the secret S is:

$$S = P(0) = \sum_{j=1}^t y_j L_j(0)$$

- where:

$$L_j(0) = \prod_{\substack{m=1 \\ m \neq j}}^t \frac{-x_m}{x_j - x_m}$$

- **Exercise:** pick any 3 shares and show that the secret s can be recovered.

Multisig and Threshold Signatures

- Multisig:
 - Simply generate multiple keys instead of one—one per device/party.
 - To authorize a transaction, all devices (or t of them) need to sign.
 - If collective signing is used, these multi-signatures can be aggregated into one signature under an aggregated public key.
- Threshold signatures:
 - A signing key (that corresponds to one verification/public key) is divided among several parties such that any t subset of them can jointly produce a signature (each produces a partial signatures such that t partials signatures can be combined into one signature that will be verified under the public key).
 - There is an increasing interest of threshold signatures these days across a large number of cryptocurrency systems (e.g., the BLS scheme).

Online Wallets

- A wallet in the form of a web app.
 - The site stores keys.
 - The site issues transactions when the users asks for that.
 - The user logs in using some credentials.
- Great usability; log in using any device connected to the Internet.
- Site compromised, wallet is compromised.
- Usually used when trading on exchanges.

Cryptocurrency Exchanges

- Pretty much a centrally managed bank system.
 - Accept deposits in cryptocurrency or fiat currency.
 - With a promise to pay back when a client ask for any withdrawal.
 - Allow customers to pay in cryptocurrency, receive payments, and trade cryptocurrencies.
 - Mainly match buyers with sellers at the exchange rate along with charging fees.
 - Nothing goes to the network, it is just that the exchange makes a different promise to the customers.
 - Everything is local.

Cryptocurrency Exchanges - Issues

- Highly convenient.
 - Several services at the same place.
 - Very close to traditional banking systems already in use.
 - Gives a direct value for cryptocurrency in terms of fiat currency.
- High risk.
 - Requires pre-identification - know your customer policy.
 - Any anonymity aspect promised by a cryptocurrency is taken away!
 - Security risks - a target for attackers.
 - Exchanges accumulate currency from all of their customers - the decentralization aspect is taken away!
- Regulation issues.
 - Traditional banks have to prove holding a specific fraction of money in reserve.

